

Hence, fraction of energy lost = $1 - 1/10 = 9/10$

16 Ans: B

Amplitude of wave from S_1 after passing polariser = $A \cos 60^\circ$

Amplitude of wave from S_2 after passing polariser = $A \cos 30^\circ$

Amplitude of resultant wave at P (vector sum) = $\text{sqrt} [(A \cos 60^\circ)^2 + (A \cos 30^\circ)^2] = A$

Hence, intensity is also I .

17 Ans: D